

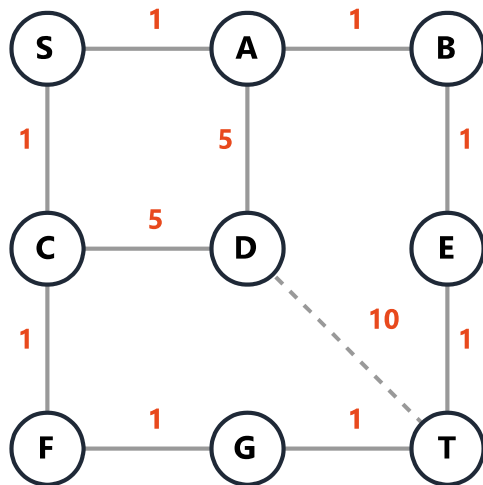


Select Lab Module:

Part 3: Student Work (Treasure Hunt) ▼

Reset Module

The Treasure Map



Heuristic Table: $h(n)$

Node	S	A	B	C	D	E	F	G	T
$h(n)$	4	3	2	3	2	1	2	1	0

Treasure Hunt Worksheet

Analyze the map and calculate the final path to the Treasure (T) chosen by each algorithm. Assume ties are broken alphabetically.

1. BFS Path (Shortest Hops)

Format: S-X-Y-T

S-A-E

2. DFS Path (Deepest First)

Format: S-X-Y-T

S-A-E

3. A* Search Path (Optimal)

Format: S-X-Y-T

S-A-E

4. Total Cost of A*

Path

4

Verify Answers



All Answers Correct!

- **BFS** fell into the Swamp trap because it blindly searches for fewest hops (3 edges).
- **DFS** luckily found an optimal path due to alphabetical tie-breaking, but it's not guaranteed.

- **A*** perfectly calculated the true costs and bypassed the high-cost Swamp.